# Zhexiang (Bruce) Li

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## EDUCATION

**University of California, Los Angeles** Sep 2026 – Jun 2028 (expected)

M.S. in Computer Science

**University of Southern California** May 2026

B.S. in Computer Science, *Magna Cum Laude* · GPA 3.76 / 4.00

## RESEARCH EXPERIENCE

**Graduate Researcher | Health Intelligence Lab (HAIL), UCLA** Aug 2026 – Present

Advisor: Prof. Yuzhe Yang

- Investigate multimodal foundation models for diabetes, integrating continuous glucose monitoring with insulin, meals, activity, and patient context.
- Curated a unified CGM data release from 49 datasets (7,000+ source person records) with meal, insulin, and wearable context, provenance checksums, device-clock fixes, and 14 automated release checks.
- Built patient-disjoint MetaboNet benchmarking and audit pipelines. Evaluated GluFormer, T1DFormer, and custom predictive-state and Transformer architectures at 30/60/120-minute forecasting horizons.

**USC Drug Discovery Research** Spring 2026

Research with a Ph.D. student in Prof. Yan Liu's group · Project poster

- Studied SARS-CoV-2 antiviral discovery through reproducible benchmarks of molecular descriptor-based models, Chemprop, and hybrid approaches across seven Therapeutics Data Commons datasets, using scaffold splits.
- Extra Trees on RDKit descriptors had the best average AUROC across datasets (85.9) and reached 90.0 ± 6.4 on SARS-CoV-2 3CLPro; analyzed descriptor importance and SHAP attributions to interpret its predictions.

## PUBLICATIONS & MANUSCRIPTS

**MedTokenBudget: Lesion-Preserving Token Routing for Dermoscopic Image Classification** 2026

**Zhexiang Li** (sole author). Under review

Developed lesion-supervised post-backbone token routing. At a 30% token budget on ISIC 2019, achieved 80.38 ± 0.18% accuracy versus 76.93 ± 0.03% for ToMe across three seeds; the frozen backbone remains unchanged.

**Mass-Conservation Token Transformers for Auditable MS/MS Spectrum Retrieval** 2026

Under review at AAAI 2027

Represented mass-conserving fragment pairs as sparse evidence tokens for spectrum retrieval. Used closest-mass and overlap-hard negatives, alongside evidence-removal tests, to examine whether the exported evidence affects ranking.

**Efficient Attention via Pre-Scoring: Prioritizing Informative Keys in Transformers** 2025

**Zhexiang Li**, Haoyu Wang, Yutong Bao, David Woodruff. Under review · arXiv:2505.11040

First-author work advised by Prof. David P. Woodruff (CMU). Proposed pre-scoring and implemented informative-key selection in Triton. Integrated with HyperAttention, reduced ChatGLM perplexity from 12.0 to 9.5 at 131k-token contexts under a fixed interaction budget.

### Game Theory in Companies 2023

S. Li, **Z. Li**, Z. Liu. *Advances in Economics, Management and Political Sciences*, 16(1), 357–366. DOI

Advised by Prof. Ming Gu (UC Berkeley). Led a team modeling game-release decisions as two-player normal-form games and enumerating Nash equilibria with Python and Gambit.

## PROFESSIONAL EXPERIENCE

### AI Engineer Intern | PMAOS / Next Big Seek Jul – Aug 2026

- Designed short-video recommendation using bidirectional user-video interest propagation, exposure budgets across multiple timescales, and fatigue controls to manage repeated recommendations.
- Developed 11-language audio-visual affect inference with evidence fusion and a fixed-point decision layer that can abstain or report conflicting modalities; implemented in C for Cortex-M3 and ARM926EJ-S targets.
- Screened sub-10 mm piezoelectric robot architectures and specified a PZT-steel cantilever experiment for a 9 mm candidate; work concluded at experiment and procurement planning.

### Full-Stack Engineer | Terlingua Threads, Remote Jan – May 2026

- Built Shopify-integrated pages and Google Maps features; improved responsive layouts, accessibility, SEO, and maintainable content structures.

### Co-Founder & CTO | CrAIg Capital, Los Angeles, CA 2025 – Mar 2026

- Led product and engineering for AI-assisted housing debt collection, including data pipelines and automated workflows.
- Set technical direction, hiring priorities, and delivery standards during startup development.

### AI Engineer Intern | Piezolab Medical, Lake Forest, CA Jun – Aug 2025

- Built an LLM-assisted pipeline to ingest, clean, and cluster 5,000+ medical-malpractice documents for quality-of-care review.
- Developed agent workflows, experiment checklists, and acceptance tests for vascular-care research.

### Business Development Intern | Hangzhou Haoan Jianke Technology, Hangzhou Jun – Aug 2024

Gathered requirements for a NetEase CodeWave internal platform, helped recruit engineers, and contributed to program structure and interface planning.

### Application Engineer Intern | Synaptics, San Jose, CA Jun – Aug 2023

Validated product firmware and built an ESP32-to-Alibaba Cloud pipeline that published I²C sensor data as MQTT topics for internal dashboards.

### Investment Analyst Intern | GUDU Medical, Remote Jan 2021 – Present

Supported MedTech deal screening, market analysis, due diligence, financial modeling, and valuation.

## SELECTED PROJECTS

### Fridge Manager 2026

Next.js · TypeScript · Supabase · LLM and vision models · howtoeat.food

- Built and launched an AI kitchen web app that plans daily and weekly meals and grocery lists from each user's inventory, leftovers, preferences, cookware, and cooking time.
- Added photo-based inventory recognition and LLM recipe import with user corrections, admin AI review, and deduplication; isolated each user's data with Supabase Postgres row-level security.
- Filled the recipe library with an offline multimodal pipeline (subtitles, ASR/OCR, keyframe vision) that turned 758 cooking videos into 121 publishable recipes.

### AI Doctor WeChat 2026

Python · FastAPI · MySQL · RAG · WeChat Mini Program

- Built and deployed a FastAPI backend (110 endpoints, 26 MySQL tables) on WeChat Cloud Hosting for a clinic's Mini Program and web workspace covering clinical teaching, clinician collaboration, and patient intake.
- Designed five-role access control so LLM-organized visit records stay drafts until a doctor confirms them; added an asyncio queue that schedules LLM calls under global, per-user, and per-API limits.
- Implemented RAG with BGE embeddings, ChromaDB cosine search, and a keyword fallback.

### Investment Analysis Agent Skill Jul 2026

Python · Claude Code

Wrote an agent skill that turns a company name or pitch deck into an evidence-backed investment report across 14 dimensions; a Python QA harness gates each section on independent and primary sources, depth, and completeness.

### AI Business Card Hub Jan 2026 – Present

Swift · SwiftUI · Vision parsing · OCR · Source code

Built photo-to-contact extraction with OCR fallback, multi-photo review, bilingual records, and searchable people and company profiles.

## TEACHING

### Tutor | Introduction to Machine Learning, UCSC Spring 2023

Prof. Lise Getoor

- Held office hours and discussion sessions on logistic regression, SVMs, and tree models.
- Co-designed Jupyter Notebook assignments and contributed to the coding autograder.

### Assistant Coach | Shanghai Bosidi Education 2014 – 2018

VEX Robotics & Olympiad in Informatics

Supported robot design and algorithmic problem-solving instruction, scrimmages, and events.

## HONORS & ACTIVITIES

**University of Southern California:** *Magna Cum Laude* (May 2026); Viterbi Dean's List (Fall 2024 – Spring 2026).

**TIE Hub:** Co-Founder Speed Date volunteer (Sep 19, 2025); five-day startup hackathon participant, including prototype development, venture design, and pitching (Sep 2025).

**Noah Business School:** Entrepreneurship, financing, and Silicon Valley company visits (2025). **USC EVMA:** Member.

## TECHNICAL SKILLS

**Programming:** Python, C/C++, JavaScript/TypeScript, SQL, Java, Swift

**Machine learning:** PyTorch, Triton, scikit-learn, RDKit, Chemprop; multimodal modeling, efficient attention, recommendation, and molecular prediction

**LLM systems:** LLM and vision-model APIs, agent workflows, retrieval-augmented generation (BGE embeddings, ChromaDB), ASR/OCR pipelines

**Engineering:** FastAPI, asyncio, SQLAlchemy, MySQL, PostgreSQL (Supabase), Next.js, Docker, Linux, Git; automated testing and data pipelines

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Updated November 2026 · Full experience archive & application resumes